ZeroGate: Project Scope

ZeroGate is a Zero Trust access platform, built on NIST SP 800-207, that checks every login, web request and SSH connection against the user's identity, device health and a live risk score. Users sign in with a custom biometric passkey, reach servers only through SSH certificates that expire in minutes, and lose access the moment risk rises. Every decision is explainable, recorded in a tamper-evident log and measured for its performance cost.

In scope

Core features are required for the defense, add-ons strengthen the thesis, and stretch goals happen only if time allows.

Feature	Priority	Owning team
Passkey login (WebAuthn) and login portal	Core	Frontend, Backend & AI
Policy Engine that evaluates every request	Core	Backend & AI
Policy Administrator: sessions and mid-session kill	Core	Backend & AI
Web reverse proxy (enforcement point)	Core	Networking
SSH Certificate Authority with short-lived certificates	Core	Security & Hardware
SSH helper CLI	Core	Networking
Device posture agent: OS version, disk encryption, root or jailbreak	Core	Frontend
Rule-based risk scoring: location, time, device	Core	Backend & AI
Hash-chained audit log and verifier	Core	Security & Hardware
Admin dashboard: policies, devices, live sessions, kill	Core	Frontend
Live latency dashboard	Core	Frontend, Networking
SecureVault: store and share credentials (desktop)	Core	Frontend, Backend & AI

Feature	Priority	Owning team
Custom biometric passkey hardware (YubiKey as fallback)	Core	Security & Hardware
Just-in-time access requests and approval queue	Add-on	Backend & AI, Frontend
ML anomaly layer on top of the rules	Add-on	Backend & AI
k-of-n Shamir account recovery	Add-on	Security & Hardware, Frontend
Attack demo panel and scripted attack scenarios	Add-on	Security & Hardware, Frontend
SSH session recording (proposed)	Add-on	Security & Hardware
Policy what-if tester (proposed)	Add-on	Backend & AI, Frontend
Single Packet Auth	Stretch	Networking
SPIFFE/SPIRE service identity	Stretch	Networking, Backend & AI

The latency dashboard and the hash-chained audit log are core rather than add-ons because they are part of what makes the project novel.

Out of scope

We deliberately do not build these, so no requirement or diagram should include them.

- The enterprise security suite: secure web gateway (SWG), CASB, data loss prevention (DLP), remote browser isolation and firewall-as-a-service. These need a global proxy network and threat intelligence.
- Access to Kubernetes clusters and databases. Only web apps and SSH servers are protected.
- Email security and AI-agent governance.
- Production deployment, real company data or real users beyond test participants.
- FIDO certification of the custom passkey.
- Hosting ZeroGate as a multi-tenant service for many organizations.

Team split

Team	Members	Owns
Security & Hardware	Abdelrehman Fouad	Custom passkey, SSH CA, audit chain, k-of-n recovery, threat model, attack scenarios
Networking	Abdelrehman Hamdy, Mohamed Elsayed	Web reverse proxy, SSH path, SSH helper CLI, Single Packet Auth, lab environment
Backend & AI	Mohamed Hashem, Dai Yasser, Rewan Fathy	Policy Engine, Policy Administrator, identity, APIs, JIT workflow, risk scoring and ML
Frontend	Mohamed Abdelwahed, Sara Mohamed, Malak Osama, Amna Waleed	Login portal, admin and latency dashboards, attack demo panel, SecureVault, posture agent